arXiv:2007.04245v1 [cs.CL] 22 Jun 2020

Understanding Object Affordances Through Verb Usage Patterns

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Abstract

In order to interact with objects in our environment, we rely on an understanding of the actions that can be performed on them, and the extent to which they rely or have an effect on the properties of the object. This knowledge is called the object "affordance". We propose an approach for creating an embedding of objects in an affordance space, in which each dimension corresponds to an aspect of meaning shared by many actions, using text corpora. This embedding makes it possible to predict which verbs will be applicable to a given object, as captured in human judgments of affordance. We show that the dimensions learned are interpretable, and that they correspond to patterns of interaction with objects. Finally, we show that they can be used to predict other dimensions of object representation that have been shown to underpin human judgments of object similarity.

1 Introduction

In order to interact with objects in our environment, we rely on an understanding of the actions that can be performed on them, and their dependence (or effect) on properties of the object. Gibson [1] coined the term "affordance" to describe what the environment "provides or furnishes the animal". Norman [2] later developed the term to focus on the properties of objects that determine the action possibilities. The notion of "affordance" therefore emerges from the relationship between the properties of objects and human actions. If we consider "object" as meaning anything concrete that one might interact with in a typical environment, there will be thousands of possibilities, both animate and inanimate (see WordNet [3]). Likewise, if we consider "action" as meaning any verb that might be applied to the noun naming an object, there will be thousands of possibilities (see VerbNet [4]). Intuitively, only a relatively small fraction of all possible combinations of object and action will be plausible. Of those, many will also be trivial, e.g. "see" or "have" may apply to almost every object.

Our first goal is determining, given an action and an object, whether the action can apply to the object. Additionally, we would like to be able to obtain most of the non-trivial actions that could be applied to a given object. This problem has been defined as "affordance mining" [5]. Our second goal is to understand what properties of objects are key to those affordances. It is possible to elicit properties of objects by asking human subjects; many of these clearly drive affordances, and are shared by many objects But how can we determine which properties are relevant to affordances?

In this paper, we will introduce a systematic approach for achieving both of these goals. Our approach is based on the hypothesis that, if a set of verbs apply to the same objects, they apply for similar reasons. We start by identifying applications of action verbs to nouns naming objects, in large text corpora. We then use the resulting dataset to produce an embedding that represents each object as a vector in a low-dimensional space, where each dimension groups verbs that tend to be applied to similar objects. We do this for larger lists of objects and action verbs than previous studies (thousands

Preprint. Under review.

in each case). We show that this allows us to predict which verbs will be applicable to a given object, as captured in human judgments of affordance. Further, we show that the dimensions learned are interpretable, and that they correspond to patterns of interaction with objects. Finally, we show that they can be used to predict other dimensions of object representation that have been shown to underpin human judgments of object similarity, and are also at the root of affordance.

2 Related Work

The authors of [5] proposed complementary methods for solving the affordance mining problem, both of which predicted a plausibility score for each combination of object and action. The best method used word co-occurrences in two ways: n-gram counts of verb-noun pairs, or similarity between verb and noun vectors in Latent Semantic Analysis [6] or Word2Vec [7] word embeddings. For evaluation, they collected Amazon Mechanical Turk (AMT) judgements of plausibility ("is it possible to <verb> a <object>") for every possible combination of 91 objects and 957 visualizable action verbs. The authors found they could retrieve a small number of affordances for each item, but precision dropped quickly with a higher recall. With the goal of understanding the relevant visual features in an object that are indicative of affordances, [8] trained a convolutional neural network (CNN) to successfully predict affordances from images. They collected affordance judgments on AMT ("what can you do with <object>") for 500 objects and harmonized them with WordNet synsets for 334 action verbs. Finally, [9] introduced a public resource of manually disambiguated lexical-semantic combinations. They first extracted pairs of co-occurring words from a large corpus, as determined from dependency parses. The resulting 78,000 word pairs were then curated by human annotators, who labelled each word with a WordNet synset. This yielded 61,249 noun-verb combinations (14,204 objects, 6,422 verbs related to them). We will use the human judgments in these datasets as ground truth for evaluating our approach, as described in Section 4.1.

There are several studies focused on obtaining properties of objects. In the two most comprehensive ones [10, 11] subjects were asked to list binary properties for hundreds of objects. Their answers were harmonized and tallied, yielding ~ 2500 and ~ 5000 properties, respectively. Properties could be taxonomic (category), functional (purpose), encyclopedic (attributes), or visual-perceptual (appearance), among other groups. While some of these are affordances in themselves (e.g. "edible"), others might suggest many affordances at once (e.g. "is a vegetable" means that it can be planted, cooked, sliced, etc). More generally, it was not clear how to rate properties by how critical they are to each object, or fundamental to human judgements about them (e.g. object similarity). This problem was tackled by [12], who proposed SPoSE, an embedding for objects derived from behavioural data, where each dimension was also interpretable (by being constrained to be sparse and positive). The embedding was learned from a database of 1.5M judgments collected in AMT, where subjects were asked "which of a random triplet of objects is the odd one out". Their goal was to predict the behaviour based solely on the similarity of the embedding vectors for the three objects. They showed that these dimensions are predictable as a *combination* of elementary properties in the [11] norm, so they correspond to properties that often co-occur across many objects. In [13], the authors further showed that 1) human subjects could coherently label what the dimensions are "about", ranging from categorical (e.g. animate, food, drink, building) to functional (e.g. container, tool) or structural (e.g. made of metal or wood, has inner structure). Subjects could predict what dimension values new objects would have, based on knowing the dimension value for a few other objects. We will use SPoSE embeddings to interpret our embedding dimensions, as described in Section 4.2.

3 Data and Methods

3.1 Word Lists

In this paper, we use the list of 1854 object concepts introduced in [13]. This sampled systematically from concrete, picturable, and nameable nouns in American English, and was further expanded by a crowdsourced study to elicit category membership ratings. Note that the list includes many things besides small objects, e.g. animals, insects, vehicles. We will refer to the objects and the nouns naming them interchangeably. We created our own verb list, by having three annotators go through the list of all verb categories on VerbNet [4], and selecting those that included verbs that corresponded to an action performed by a human towards an object. We kept all the verbs in each selected category, and selected only those categories where all annotators agreed. The resulting list has 2541 verbs.

3.2 Extraction of Verb Applications to Nouns from Text Corpora

We used the UKWaC and Wackypedia corpora [14] containing, approximately, 2B and 1B tokens, and 88M and 43M sentences, respectively. The former is the result of a crawl of British web pages, while the latter is a subset of Wikipedia. Both have been extensively cleaned up and have clearly demarcated sentences, which makes them ideal for dependency parsing. We identified all 26M sentences containing both verbs and nouns in our list, and we used Stanza [15] to produce dependency parses for them. We extracted all the noun-verb pairs in which the verb was a syntactic head of a noun having `obj` (object) or `nsubj:pass` (passive nominal subject) dependency relations. We compiled raw counts of how often each verb was used on each noun, producing a count matrix M. Note that this is very different from normal co-occurrence counts - those would register a count whenever verb and noun were both present within a short window (e.g. up to 5 words away from each other), *regardless* of whether the verb applied to the noun, or they were simply in the same sentence. Out of the 1854 nouns considered, there were 1530 with at least one associated action, and this is the list we will use in the remainder of this paper. Note also that the counts pertain to every possible meaning of the noun, given that no word sense disambiguation was performed.

Finally, we converted the matrix M into a Positive Pointwise Mutual Information (PPMI [16]) matrix P where, for each object i and verb k:

$$P(i,k) := \max\left(\log \frac{\mathbb{P}(M_{ik})}{\mathbb{P}(M_{i*}) \cdot \mathbb{P}(M_{*k})}, 0\right), \tag{1}$$

where $\mathbb{P}(M_{i*})$ and $\mathbb{P}(M_{*k})$ are respectively the marginal probability of i and k. P can be viewed as a pair-pattern matrix [17], where the PPMI helps in separating frequency from informativess of the co-occurrence of nouns and verbs [16, 18]. However, PPMI is also biased and may be large for rare co-occurrences, e.g. for (o_i, v_k) that co-occur only once in M. This is addressed in the process of generating an embedding through factorizing P, as described in the next section.

3.3 Object embedding

Object embedding via matrix factorization Our object embedding is based on a factorization of the PPMI matrix P (m objects by n verbs) into the product of low-rank matrices O (m objects by d dimensions) and V (n verbs by d dimensions), yielding an approximation $\widetilde{P} := OV^T \approx P$. Intuitively, if two verbs occur often with the same objects, they will both have high loadings on one of the d-dimensions; conversely, the objects they occur with will share high loadings on that dimension. O is the object embedding in d-dimensional space, and V is the verb loading for each of the d dimensions. The idea of factoring a count matrix (or a transformation thereof) dates back at least to Latent Semantic Analysis [19], and was subsequently investigated by many others ([16] is a good review). Given that PPMI is biased towards rare pairs of noun/verb, the matrix P is not necessarily very sparse. If factorized into a product of two low-rank matrices, however, the structure of the matrix can be approximated while excluding noise or rare events [20].

Optimization problem Given that PPMI is positive, the matrices O and V are as well. This allows us to obtain them through a non-negative matrix factorization (NMF) problem

$$O^*, V^* = \underset{O,V}{\text{argmin}} \|P - OV^T\|_F^2 + \beta\mathcal{R}(O,V), \tag{2}$$

which can be solved through an iterative minimization procedure. For the regularization $\mathcal{R}(O,V)$, we chose the sparsity control $\mathcal{R}(O,V) \equiv \sum_{ij} O_{ij} + \sum_{ij} V_{ij}$.

In our experiments, we use $d = 70$ and $\beta = 0.3$. These values were found using the two-dimensional hold-out cross validation proposed in [21], due to its scalability and natural fit to the multiplicative update algorithm for solving (2). Specifically, denoting M_t and M_v to be the mask matrices representing the held-in and held-out entries, we optimize for

$$O^*, V^* = \underset{O,V}{\text{argmin}} \|M_t \odot (P - OV^T)\|_F^2 + \beta\mathcal{R}(O,V) \tag{3}$$

and obtain the reconstruction error $E = \|M_v \odot (P - O^*(V^*)^T)\|_F^2 + \beta\mathcal{R}(O^*, V^*)$. For more details, please refer to the Supplementary Materials.

The optimization problem (2) is NP-hard and all state-of-the-art algorithms may converge only to a local minimum [22]. Hence, choosing a proper initialization of O and V is crucial. We used the NNDSVD initialization [23], a SVD-based initialization which favours sparsity on O and V and approximation error reduction.

Estimating the verb usage pattern for each object Each column $V_{:,k}$ of matrix V contains a pattern of verb usage for *dimension* k, which captures verb co-occurrence across all objects. Deriving a similar pattern for each *object* i, given its embedding vector $O_{i,:} = [o_{i_1}, o_{i_2}, \dots o_{i_d}]$, requires combining these patterns based on the weights given to each dimension. The first step in doing so requires computing the cosine similarity between each embedding dimension $O_{:,h}$ and the PPMI values $\tilde{P}_{:,k}$ for each verb k in the approximated PPMI matrix $\tilde{P} = OV^T$, which is

$$S(O_{:,h}, \tilde{P}_{:,k}) = \frac{O_{:,h} \cdot \tilde{P}_{:,k}}{\|O_{:,h}\|_2 \|\tilde{P}_{:,k}\|_2}. \tag{4}$$

Given the embedding vector for object i, $O_{i,:} = [o_{i_1}, o_{i_2}, \dots o_{i_d}]$, we compute the pattern of verb usage for the object as $O_{i,:}S$. Thus, this is a weighted sum of the similarity between every $O_{:,h}$ and $\tilde{P}_{:,k}$. We will refer to the ordering of verbs by this score as the *verb ranking* for object i.

4 Experiments and Results

4.1 Prediction of affordance plausibility

Affordance ranking task The first quantitative evaluation of our embedding focuses on the ranking of verbs as possible affordances for each object. We will use the Affordance Area Under The Curve (AAUC) relative to datasets that provide, for each object, a set of verbs known (or likely) to be affordances. Intuitively, the verb ranking for object i is good if it places these verbs close to the top of the ranking, yielding an AAUC of 1. Conversely, a random verb ranking would have an AAUC of 0.5, on average. Note that this is a conservative measure, given that the set of possible affordances for each object is much larger than those in the datasets. However, it is useful for comparing between our and competing approaches for producing rankings. More formally, given the K ground truth verb affordances $\{g_k\}_{k=1}^K$ of object i, and its verb ranking $\{v_i\}_{i=1}^n$, we denote ℓ_k to be the index such that $v_{\ell_k} = g_k \; \forall k$. We then define AUCC for object i as AUCC $= \frac{1}{K} \sum_{k=1}^{K} \left(1 - \frac{\ell_k}{n}\right)$.

Datasets We used two different datasets as ground truth for the set of known affordances for each object. In the first dataset (WTAction [8]), objects are associated with the top 5 actions label provided by human annotators in response to "What can you do with this object?". Out of 1,046 objects and 445 actions in this dataset, there are 342 objects and 323 verbs that overlap with ours. On average, there are 3.13 action labels per object in the resulting dataset. The second dataset (SyntagNet [9]) contains noun-verb combinations extracted from a combination of the English Wikipedia and the British National Corpus [24]. There are 52,432 noun-verb relations covered in SyntagNet, including `compound`, `dobj` (direct object), `iobj` (indirect object), `nsubj` (nominal subject) and `nmod` (nominal modifier). Out of 7,669 nouns and 3,786 verbs in this dataset, there are 712 nouns and 1,067 verbs that overlap with ours. On average, there are 7.17 verbs per object in the resulting dataset. Note that these are *not* just affordance judgments, but rather any way in which a verb and an object could relate (e.g. entailment or property of the object). We considered using the [5] dataset as a third benchmark but opted not to, given that it is much smaller than these two in the number of objects considered.

Baseline methods We compared the ranking of verbs produced by our algorithm with an alternative proposed in [5]: ranking by the cosine similarity between pre-existing embedding vectors for nouns and those for all possible verbs. We considered three embedding alternatives, Word2Vec [7] (300D, trained on a 6B token corpus) and GloVe [25] (300D, two versions, trained on 6B and 42B token corpora). We contrasted Word2Vec and GloVe because they are based on two different embedding approaches (negative sampling and decomposition of a word co-occurrence matrix), developed on corpora twice as large as ours. We contrasted 6B and 42B versions of GloVe to see the effect of vastly increasing dataset size. In both methods, the co-occurrence considered is simply proximity within a window of a few tokens, rather than application of the verb to the noun. Finally, we also ranked the verbs by their values in the row of the PPMI matrix P corresponding to an object, to see the effect of using a low-rank approximation in extracting information.

Table 1: Average AAUC of verb rankings produced by our and baseline methods

ground truth dataset	verb ranking method				
	Ours	Word2Vec	GloVe 6B	GloVe 42B	PPMI
WTAction	**0.90**	0.74	0.80	0.84	0.80
SyntagNet	**0.90**	0.81	0.83	0.86	0.83

Results For both datasets, we reduced our embeddings O and V according to the sets of objects and verbs available. We then obtained the verb ranking for each object, as described in Section 3.3, as well as rankings predicted with Word2Vec and GloVe embedding vector similarity, and the values in the PPMI matrix P. Table 1 shows the AAUC results obtained with these verb rankings on the two datasets. Our ranking is better than that of all three baseline methods, as determined from a paired two-sided t-test, in both WTAction (p-values of $1.33\text{e}{-47}$, $9.68\text{e}{-29}$, and $2.28\text{e}{-17}$, respectively) and SyntagNet (p-values of $8.34\text{e}{-46}$, $4.93\text{e}{-23}$, and $2.04\text{e}{-09}$, respectively) datasets.

The overall distributions of AAUCs for the two datasets are shown in Figure 1. Our procedure yields AAUC closer to 1.0 for many more items than the other methods. This suggests that the co-occurrence of verb and noun within a window of a few tokens, the basis of the word embeddings that we compare against, carries some information about affordances, but also includes other relationships and noise (e.g. if the verb is in one clause and the noun in another). Results are better in the embedding trained in a corpus 14X larger, but still statistically worse than those of our procedure (it performs better on SyntagNet, where some of the noun/verb interactions are frequent, but not affordances). Ranking based on the PPMI matrix P performs at the level of the 6B token embeddings. This suggests that our procedure is effective at removing extraneous information in the matrix P.

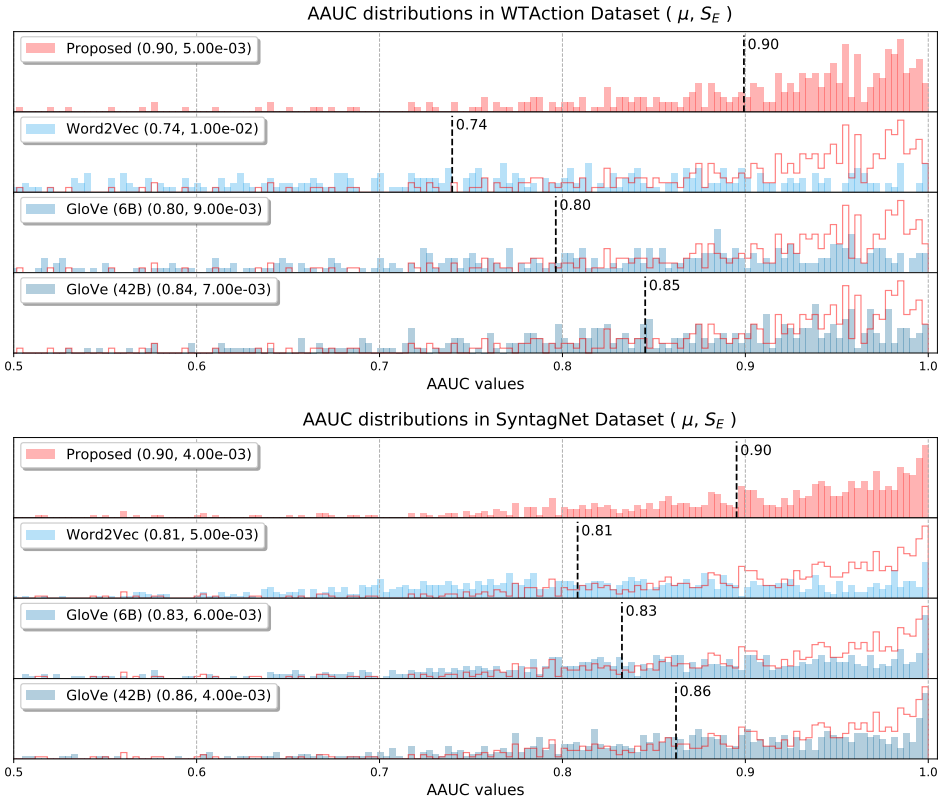


Figure 1: **AAUC performance of our method versus Word2Vec and GloVe (6B and 42B) baselines, in two datasets.** The AAUC distributions over the 342 objects in WTAction dataset (top), and the 712 objects in SyntagNet (bottom) are shown, with the distribution of our method overlaid on baselines. The mean and the standard error (μ, S_E) of the distributions are in the legend of the figure.

4.2 Relation between object properties and affordances

The SPoSE representation and dataset The dimensions in the SPoSE representation [13] are interpretable, in that human subjects coherently label what those dimensions are "about", ranging from the categorical (e.g. animate, building) to the functional (e.g. can tie, can contain, flammable) or structural (e.g. made of metal or wood, has inner structure). Subjects can also predict what dimension values new objects would have, based on knowing the dimension value for a few other objects. The SPoSE vectors for objects are derived solely from behaviour in a "which of a random triplet of objects is the odd one out" task. The authors propose a hypothesis for why there is enough information in this data to allow this: when given any two objects to consider, subjects mentally sample the contexts where they might be found or used. The resulting dimensions reflect the aspects of the mental representation of an object that come up in that sampling process. The natural question is, then, which of these dimensions drive affordances, and that is what our experiments aim to answer.

For our experiments, we used the 49-D SPoSE embedding of 1854 unique objects published with [13]. Out of these, we excluded objects named by nouns that had no verb co-occurrences in our dataset and, conversely, verbs that had no interaction with any objects. We averaged the vectors for each set of objects named by the same polysemous noun (e.g. "bat"). This resulted in a dataset of 1530 objects/nouns, with their respective SPoSE embedding vectors and 2451 verbs.

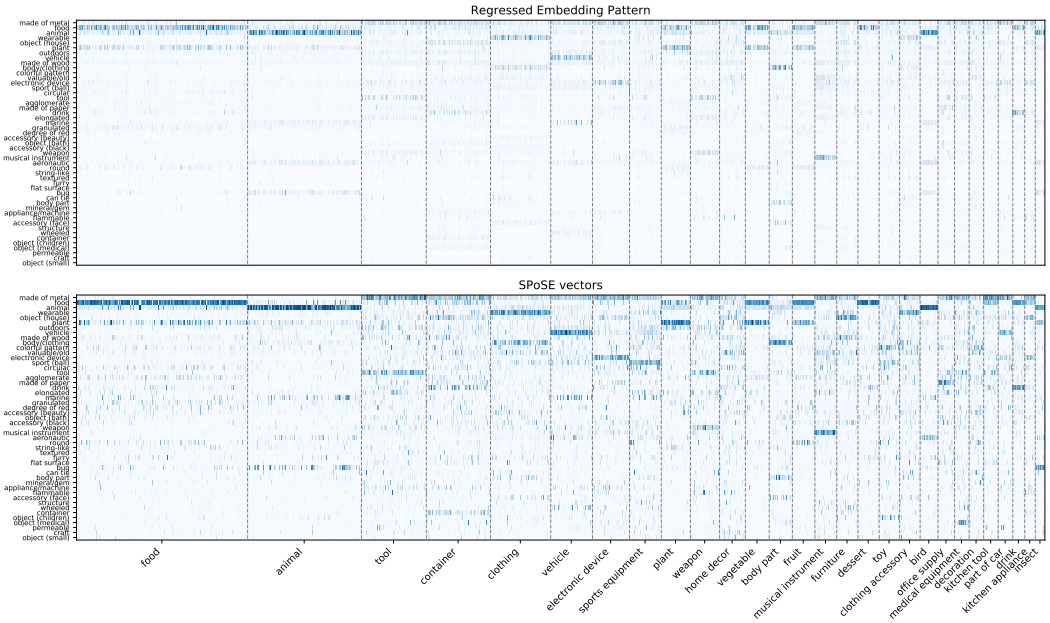


Figure 2: **Prediction of SPoSE embeddings from affordance embeddings (top) versus actual SPoSE embeddings (bottom).** Objects are grouped by semantic category (those with ≥ 15 objects).

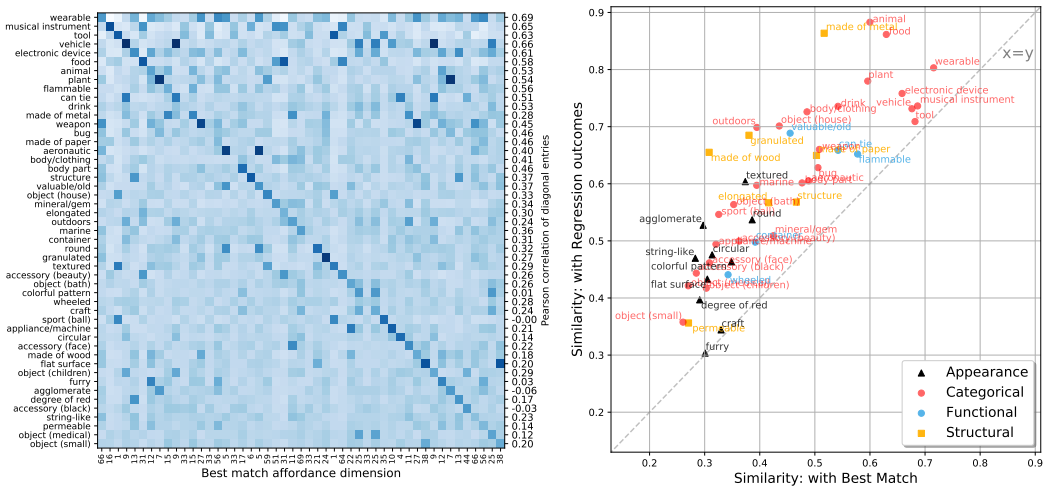
Prediction of SPoSE meaning dimensions from affordance embedding The first experiment we carried out was to predict the SPoSE dimensions for an object from its representation in our affordance embedding. The ability to do this tells us which SPoSE dimensions – and, crucially, which *kinds* of SPoSE dimensions – can be explained in terms of affordances, to some extent.

Denoting the SPoSE embeddings for m objects as a $m \times 49$ matrix Y, we solved the following ridge regression problem for each column $Y_{:,i}$

$$w_i^* = \arg \min_{w \in \mathbb{R}^d, w \geq 0} \frac{1}{2m} \|Y_{:,i} - Ow\|_2^2 + \lambda \|w\|_1, \quad i = 1, 2, \ldots 49, \tag{5}$$

where λ was chosen based on a 2-Fold cross-validation, with λ ranging from $[0, 10]$ with log-scale spacing. Since both $Y_{:,i}$ and our embedding O represent object features by positive values, we restricted $w \geq 0$. Intuitively, this means that we try to explain every SPoSE dimension by combination of the *presence* of certain affordance dimensions, not by trading them off.

Figure 2 contrasts the predictions $\tilde{Y}_{:,i} = Ow_i^*$ and true values $Y_{:,i}$ for each SPoSE dimension i. Objects are grouped by their semantic category. To keep the figure uncluttered, we only plotted loading of objects in categories with size ≥ 15. The visual resemblance of the patterns, and the range of correlations between true and predicted dimensions (0.3-0.88, see Figure 3b for distribution), indicate that the affordance embedding contains enough information to predict most SPoSE dimensions.



(a) Correlation with outcomes

(b) Comparison

Figure 3: **a)** Correlation between each SPoSE dimension and the corresponding best match in our affordance embedding (correlation values shown in right vertical axis). **b)** For each SPoSE dimension, the relationship between the similarity with the best matching affordance dimension (x-axis) and the similarity with the cross-validated prediction of the regression model for it (y-axis).

Relationship between SPoSE and affordance dimensions We first considered the question of whether affordance dimensions correspond directly to SPoSE dimensions, by looking for the best match for each of the latter in terms of cosine similarity. As shown in Figure 3(a), most SPoSE dimensions have one or more affordance dimensions that are similar to them. However, when we consider the cross-validated regression models to predict SPoSE dimensions from affordance dimensions, *every* model places non-zero weight on several of the latter. Furthermore, those predictions are much more similar to SPoSE dimensions than almost any individual affordance dimension. Figure 3(b) plots, for every SPoSE dimension, the correlation with its closest match (x-axis) versus the correlation with the cross-validated prediction of the regression model (y-axis). The best predicted dimensions are categorical, e.g. "animal", "plant", or "tool", or functional, e.g. "can tie" or "flammable". Structural dimensions are also predictable, e.g. "made of metal", "made of wood", or "paper", but appearance-related dimensions, e.g. "colorful pattern", "craft", or "degree of red", less so.

What can explain this pattern of predictability? Most SPoSE dimensions can be expressed as a combination of the presence of affordance dimensions. Each affordance dimension, in turn, corresponds to a ranking over verbs. If we consider the top 5 verbs from affordance dimensions that are most used in predicting each SPoSE dimension, we see that "tool" has "wield, sharpen, grab, hold, swing" (D5); "animal" has "spot, observe, photograph, find, sight" (D1), "hunt, kill, stalk, cull, chase" (D3), or "pasture, slaughter, milk, herd, bleat" (D19); "food" has "eat, munch, serve, bake, scoff" (D16), or "boil, blanch, cook, steam, drain" (D2); "plant" shares D2 with "food", but also has "cultivate, grow, plant, prune, propagate" (D31). The full list of affordance dimensions for each SPoSE dimension is provided in Supplementary Materials Table 6–9.

These results suggest that SPoSE dimensions are predictable *insofar* as they can be expressed as combinations of modes of interaction with objects. Using the approach in Section 3.3, we produced a combined ranking over verbs for each SPoSE dimension. We first replaced the embedding O in (4) with the SPoSE prediction $\widetilde{Y}$ and we ranked the verbs for dimension h according to $S(\tilde{Y}_{:,h}, \tilde{P}_k)$. Table 2 shows, for every SPoSE dimension, ranked by predictability, the top 10 verbs in its ranking. Examining this table provides empirical confirmation that, first and foremost, highly

predictable categorical dimensions correspond to very clean affordances. The same is true for functional dimensions, e.g. "can tie" or "container" or "flammable"; even though they are not "classic" categories, subjects group items belonging to them based on their being suitable for a purpose (e.g. "fasten", "fill", or "burn"). But why would this hold for structural dimensions? One possible reason is that objects having that dimension overlap substantially with a known category (e.g. "made of metal" and "tool"). Another is that the structure drives manual or mechanical affordance (e.g. "elongated" or "granulated"). Finally, what are the affordances for appearance dimensions that can be predicted? Primarily, actions on items in categories that share that appearance, e.g. "textured" is shared by fabric items, "round" is shared by many fruits or vegetables. Prediction appears to be worse primarily when the items sharing the dimension are very heterogeneous in terms of belonging to multiple categories (judged with the THINGS [26] database, which contains the pictures of items shown to subjects).

Pearson correlation	Dimension label	Taxonomy	Affordances (Top Ten Ranked Verbs)
0.83	animal	categorical	kill, hunt, spot, chase, observe, feed, slaughter, find, sight, photograph
0.79	food	categorical	eat, serve, cook, taste, consume, prepare, add, mix, munch, sell
0.76	wearable	categorical	wear, don, embroider, knit, sew, fasten, match, rip, button, tear
0.69	plant	categorical	grow, cultivate, plant, chop, eat, add, boil, mix, sow, cut
0.67	made of metal	structural	fit, invent, manufacture, incorporate, design, utilize, attach, position, mount, fix
0.62	tool	categorical	wield, grab, sharpen, hold, carry, swing, hand, grasp, clutch, thrust
0.61	flammable	functional	light, extinguish, ignite, kindle, rekindle, douse, fuel, burn, flame, flare
0.59	can tie	functional	fasten, unfasten, tighten, undo, thread, unclip, attach, loosen, slacken, secure
0.56	granulated	structural	deposit, mix, sprinkle, contain, shovel, quarry, scatter, mine, extract, grind
0.49	textured	appearance	weave, spread, drape, stitch, fold, unfold, stain, tear, traverse, hang
0.42	round	appearance	cultivate, grow, slice, peel, chop, plant, grate, mash, pick, cut
0.43	made of wood	structural	place, incorporate, construct, remove, design, carry, carve, manufacture, fit, situate
0.38	elongated	structural	wield, swing, hold, grab, carry, grip, grasp, sharpen, attach, thrust
0.39	container	functional	empty, fill, unload, load, stack, clean, dump, lug, carry, stow
0.23	colorful pattern	appearance	weave, wrap, sew, hang, drape, spread, unfold, embroider, apply, stitch
0.21	permeable	structural	fit, incorporate, position, mount, block, rotate, install, fix, utilize, attach
0.18	degree of red	appearance	cultivate, grow, plant, prune, dry, propagate, gather, combine, mix, pick

Table 2: Affordance assignment for SPoSE dimensions mentioned in the text, ordered by how well they can be predicted from the affordance embedding. Dimension labels are simplified. The complete table and the detailed descriptions for each dimension are provided in Supplementary Materials Table 3, Table 4, and Table 5, respectively.

5 Conclusions

In this paper, we introduced an approach to embed objects in a space where every dimension corresponds to a pattern of verb applicability to those objects. We showed that this embedding can be learned from a text corpus and used to rank verbs by how applicable they would be to a given object. We evaluated this prediction against two separate human judgment ground truth datasets, and verified that our method outperforms general-purpose word embeddings trained on much larger text corpora. Given this validation of the embedding, we were able to use it to predict SPoSE dimensions for objects. This is an embedding that was derived from human behavioural judgements of object similarity and has interpretable dimensions, which have been validated in psychological experiments. We used the resulting prediction models to probe the relationship between affordances (as patterns of verb co-occurrence over many objects) and the various types of SPoSE dimensions. This allowed us to conclude that affordance judgements for objects are driven by 1) category information, 2) knowledge of purpose, and 3) some structural aspects of the object. To increase prediction quality in future work, one approach will be to enrich and refine the co-occurrence matrix in larger corpora, now that the basic approach has been shown to be feasible. Another interesting future direction will be to understand how affordance could be driven by more fine-grained visual appearance properties, by considering other semantic dependencies, or jointly using text and image features such as [8].

Acknowledgements

This work was supported by the National Institute of Mental Health Intramural Research Program (ZIC-MH002968, ZIA MH002035). This work utilized the computational resources of the NIH HPC Biowulf cluster (http://hpc.nih.gov). The authors would like to thank Martin Hebart and Charles Zheng for patiently sharing SPoSE and THINGS resources with us.

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Supplementary Materials: Understanding Object Affordances Through Verb Usage Patterns

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Hyper-parameter Selection for Non-negative Matrix Factorization

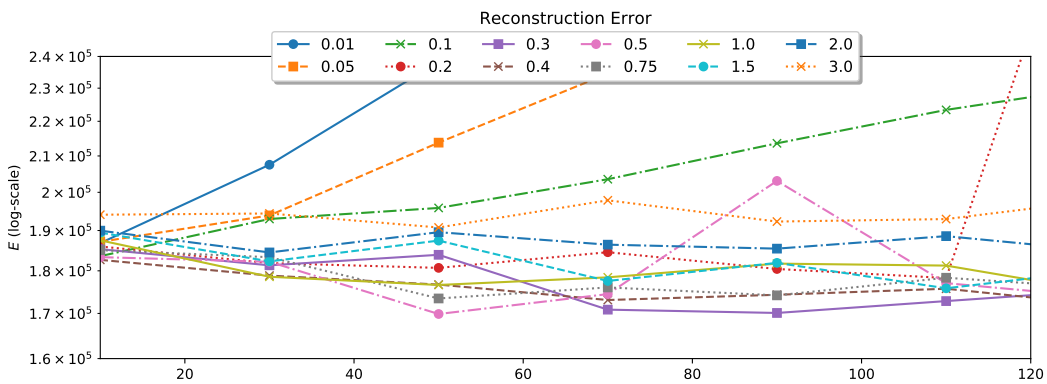

Figure 4: A zoom-in plot for the reconstruction errors.

Denote $M_t, M_v \in \{0,1\}^{n\times m}$ to be the mask matrices for indicating held-in and held-out entries of the input PPMI matrix P in CV procedure, we then optimize for O^* and V^*:

$$O^*, V^* = \underset{O,V}{\text{argmin}} \|M_t \odot (P - OV^T)\|_F^2 + \beta\mathcal{R}(O,V). \tag{6}$$

To apply the multiplicative update scheme as proposed in [27], we first consider the partial derivatives with respect to O and V. Denote $F(O,V) \equiv \|M_t \odot (P - OV^T)\|_F^2 + \beta\mathcal{R}(O,V)$, we have

$$\begin{aligned} \nabla_O F(O,V) &= (M_t \odot OV^T)V - (M \odot P)V + \beta \cdot \mathbf{1} \\ \nabla_V F(O,V) &= (M_t \odot OV^T)^T U - (M \odot P)^T U + \beta \cdot \mathbf{1}. \end{aligned} \tag{7}$$

We then have the following update rules that is guaranteed to be non-increasing:

$$\begin{aligned} O^{(i+1)} &\leftarrow O^{(i)} \odot \frac{(M_t \odot P)V^{(i)}}{(M_t \odot O^{(i)}(V^{(i)})^T)V^{(i)} + \beta} \\ V^{(i+1)} &\leftarrow V^{(i)} \odot \frac{(M_t \odot P)^T U^{(i)}}{(M_t \odot O^{(i)}(V^{(i)})^T)^T U^{(i)} + \beta}, \end{aligned} \tag{8}$$

where the fraction here represents elementary-wise division. For the choice of M_t and M_v, we follow the same approach as proposed in [21]. We first split the matrix into K blocks, where the rows and columns are randomly shuffled. Denote $\mathtt{r}^{(k)}$ and $\mathtt{c}^{(k)}$ to be the index vectors for rows and columns

respectively, where $\mathtt{r}_i^{(k)} = 1$ if row i contains in block k, or $\mathtt{c}_j^{(k)} = 1$ if column j contains in block k. The mask for k-th block can then be expressed as $M^{(k)} = \mathtt{r}^{(k)} \otimes \mathtt{c}^{(k)}$. We then randomly select q out of K blocks as holdout blocks, which gives

$$M_v = \sum_{s=1}^{q} \mathtt{r}^{(k_s)} \otimes \mathtt{c}^{(k_s)}, \quad M_t = \mathbf{1} - M_v, \tag{9}$$

where k_s is the index of selected block. The reconstruction error E can thus be computed:

$$E = \|M_v \odot (P - O^*(V^*)^T)\|_F^2 + \beta \mathcal{R}(O^*, V^*). \tag{10}$$

Figure 4 shows a zoom-in plot of the reconstruction error under different combinations of d and β. For every (d, β) setting, we perform multiple optimization since NMF is sensitive to initialization. We then choose $d = 70$ and $\beta = 0.3$ accordingly. Empirically, we observe that the rank selection is quite robust to over-fitting when there is a sufficient sparsity control, for instance, $\beta > 0.1$ in our dataset. We also observe that whenever $d \in [50, 150]$ and $\beta \in [0.05, 0.5]$, the results are similar.

Affordance assignment for each SPoSE dimension

Pearson correlation	Dimension label	Taxonomy	Affordances (Top Ten Ranked Verbs)
0.83	animal	categorical	kill, hunt, spot, chase, observe, feed, slaughter, find, sight, photograph
0.79	food	categorical	eat, serve, cook, taste, consume, prepare, add, mix, munch, sell
0.76	wearable	categorical	wear, don, embroider, knit, sew, fasten, match, rip, button, tear
0.69	plant	categorical	grow, cultivate, plant, chop, eat, add, boil, mix, sow, cut
0.70	vehicle	categorical	park, hire, commandeer, board, rent, drive, crash, equip, tow, charter
0.67	made of metal	structural	fit, invent, manufacture, incorporate, design, utilize, attach, position, mount, fix
0.66	electronic device	categorical	connect, invent, install, integrate, incorporate, design, operate, plug, disconnect, activate
0.67	body/clothing	categorical	wear, bandage, don, straighten, hurt, injure, sprain, bruise, slash, twitch
0.69	musical instrument	categorical	play, amplify, hear, sound, learn, perform, study, chime, sing, tinkle
0.62	tool	categorical	wield, grab, sharpen, hold, carry, swing, hand, grasp, clutch, thrust
0.65	drink	categorical	drink, pour, sip, swig, quaff, sup, spill, guzzle, bottle, swill
0.61	flammable	functional	light, extinguish, ignite, kindle, rekindle, douse, fuel, burn, flame, flare
0.59	can tie	functional	fasten, unfasten, tighten, undo, thread, unclip, attach, loosen, slacken, secure
0.56	bug	categorical	spot, observe, kill, find, catch, battle, chase, deter, eradicate, photograph
0.56	granulated	structural	deposit, mix, sprinkle, contain, shovel, quarry, scatter, mine, extract, grind
0.56	weapon	categorical	fire, hurl, wield, throw, carry, lob, drop, detonate, explode, grab
0.54	valuable/old	functional	carve, inscribe, craft, steal, adorn, recover, discover, decorate, pawn, hide
0.51	object (house)	categorical	fit, incorporate, empty, fill, clean, install, construct, design, place, position
0.47	aeronautic	categorical	spot, observe, sight, photograph, find, note, watch, chase, notice, kill
0.49	textured	appearance	weave, spread, drape, stitch, fold, unfold, stain, tear, traverse, hang
0.51	outdoors	categorical	construct, situate, dismantle, build, carry, repair, install, surround, erect, climb
0.53	body part	categorical	sprain, fracture, bandage, flex, hyperextend, bruise, injure, straighten, hurt, rest
0.50	made of paper	structural	print, scan, paste, annotate, archive, mail, circulate, photocopy, forward, erase
0.42	round	appearance	cultivate, grow, slice, peel, chop, plant, grate, mash, pick, cut
0.43	made of wood	structural	place, incorporate, construct, remove, design, carry, carve, manufacture, fit, situate
0.49	marine	categorical	spot, sight, beach, sail, moor, board, capsize, observe, charter, berth
0.45	structure	structural	construct, install, dismantle, erect, build, situate, fit, incorporate, mount, design
0.40	mineral/gem	categorical	mine, steal, recover, retrieve, deposit, glow, smuggle, hide, dissolve, pawn
0.39	object (bath)	categorical	clean, apply, wipe, manufacture, dab, rub, invent, remove, formulate, rinse
0.37	accessory (face)	categorical	spritz, dehair, moisturize, freckle, wear, cringe, don, discolor, flinch, crinkle
0.38	elongated	structural	wield, swing, hold, grab, carry, grip, grasp, sharpen, attach, thrust
0.39	container	functional	empty, fill, unload, load, stack, clean, dump, lug, carry, stow
0.35	accessory (beauty)	categorical	steal, wear, pawn, retrieve, recover, don, snatch, hide, smuggle, discover
0.31	wheeled	functional	hire, unload, park, drive, load, rent, crash, equip, wheel, haul
0.36	string-like	appearance	braid, cut, snip, knot, part, coil, wrap, weave, sever, loop
0.29	agglomerate	appearance	coat, soak, place, dry, contain, gather, transport, store, ship, collect
0.28	object (children)	categorical	remonstrate, conk, room, prattle, billet, swaddle, feminize, stutter, waken, shush
0.29	sport (ball)	categorical	design, manufacture, fit, position, attach, utilize, invent, incorporate, move, pull
0.31	appliance/machine	categorical	fit, install, connect, incorporate, manufacture, unload, design, equip, utilize, load
0.29	flat surface	appearance	weave, drape, wrap, unfold, fold, wet, hang, sew, wave, tear
0.25	object (medical)	categorical	invent, clean, recover, manufacture, empty, steal, hide, contain, fill, possess
0.24	craft	appearance	weave, drape, wrap, unfold, embroider, wave, sew, fold, knot, hang
0.25	circular	appearance	strangulate, ingrain, incorporate, fit, position, cover, punt, invent, place, knock
0.23	colorful pattern	appearance	weave, wrap, sew, hang, drape, spread, unfold, embroider, apply, stitch
0.21	permeable	structural	fit, incorporate, position, mount, block, rotate, install, fix, utilize, attach
0.23	accessory (black)	categorical	formulate, pour, dilute, dissolve, smear, apply, wear, squirt, spill, remove
0.23	object (small)	categorical	light, quarry, extinguish, coil, indurate, ignite, vitrify, mine, rekindle, fuse
0.18	degree of red	appearance	cultivate, grow, plant, prune, dry, propagate, gather, combine, mix, pick
0.19	furry	appearance	lour, roil, pasture, taxi, slaughter, scud, emanate, fragment, ossify, milk

Table 3: Affordance assignment for SPoSE vectors ordered in terms of Pearson correlation with regression outcomes. The dimension labels are vastly simplified. The full descriptions for each dimension are provided in Supplementary Materials Table 4 and Table 5.

Full descriptions for each SPoSE dimension

The abbreviation and the full description of every SPoSE dimension.

Abbreviation	Full descriptions
accessory (black)	accessories, beauty, black, blackness, classy, date, emphasize, fancy, hair, hard, high-class, manly, objects, picture, telescope
accessory (face)	accessories, body parts, culture, decoration, eyes, face, face accessories, facial, goes on head, hair, head, less appealing, senses, touches face, wearable
accessory (beauty)	accessory, beautiful, beauty, color, fancyness, feminine, feminine items, floral, flowers, flowery, gentle, girly, love, muted colors, pastel, pink
aeronautic	aero-nautic, air, airplanes, aviary, aviation, buoyant, flies, flight, fly, flying, flying to not, high in air, light, move, sky, swim, transportation, travel
agglomerate	accumulatable, bundles, collection, colors, countable, grainy to smooth, group of similar things, groupings, groups, groups of small objects, large groups, little bits of things, many, metals, nuts, objects, patterns, piles, quantity of objects in photo, round, small, small objects in groups, small parts that look alike, symmetrical
animal	animal, animals, animals in zoo, animals that do not fly, from complex to less, fuzzy, grass, ground animals, land animals, mammal, mammals, natural, size, wild animals, wild to human-made, wilderness
appliance/machine	building materials, construction, destructive, electric items, factory, farm tools, foundation, home tools, in groups, long, machinery, maintenance, mostly orange, processing, renovation, rocks, rope-like, thing, tool, tools
body part	body, body parts, esp extremities, extremities, extremities of body, feet, feet to hands, fingers, found on people, hand, hands, human, legs, limbs, lower body, skin
body/clothing	bodies, body, body accessories, body maintenance, body part- related, body parts, body parts with hair, face, how much skin showing, human, human body parts, part of body, parts, people, skin, touched by skin
bug	animals that stick onto things, ants, bug, bugs, can hurt you, dangerous, gardening, insects, interact with bugs, small animals, small to large, wild
can tie	bands, bondages, can tie, chained, circles, coils, construction, fasteners, knotting, long, rope, ropes, round, string-like, strings, tensile, thing can tie around, tied, ties, trapped, violent, wires, wrap, wrapped around to what gets wrapped
circular	circles, circular, cylindrical, discs, flat, round, shape, targets
colorful pattern	artistic, bright, bright colors, color, color variety, color vibrancy, colorful, colors, many colors, patterns
container	able to put something in it, boxes, buckets, can put things into, carts, container, containers, containers for liquids, containing, covering, cylinders, diverse, drums, enclosed objects, hold other things, hollow tubes, paints, shapes, storage, unknown
craft	a lot of patterns, art, artisinal, arts, candles, circles, color, crafts, detailed dots, do-it-yourself, grandma, grandparent-like, handmade, home patterns, home-making related, housework, in grandma's home, intricacy, quilt, rectangles, sewing, specks, stitching, twine, unknown, weaving, wood, woven, yarn
degree of red	color, colors, degree of redness, red, red (bright)
drink	3-dimensional, beverage, containers, containers for liquids, drinks, edible, glass, glasses, hold liquid, liquid, liquids, other things, things that fit in containers, things that hold liquids, vessels with liquids, viscosity
electronic device	digital, digital devices, digital media, electric, electronic, electronics, hard, hard to understand, media, old technology, technological, technology, telephones, typing instruments
elongated	able to be held, cane, cylinders, cylindrical, darts, grouped, long, long narrow, long objects, long-shaped, narrow, pen-like, pencils, pens, shape, sharp, skinny rectangles, stick-like, sticks, straight, straight to curved, symmetrical, thin
flammable	fire, flammability, flammable, heat, hot, light, outdoors, warm
flat surface	attaching, breakable, clean, cloth, convenience, disposable, flat coverings, gathering, grated pattern, handle everyday, helpful, hold things in, multi-shaped, not smooth to smooth, paper, paper-like, sheets, stick-like, thin, things that roll, tissue, white
food	baked food, baked goods, carbs, cheesy, comforting, cooked, deliciousness, edible, entrees, food, made dish, natural products, nutrients, pastry, prepared food, processed, salt, where it comes from
furry	fluffy, furry, more of one color, white, white and fluffy, winter
granulated	a lot of items, ash, color, dirt, elements, grain-looking, grains, grainy, grainyness, granular objects, ground, ground (grinded), homogeneity, lots of same, many, not colorful, particles, rocky, shape, size of particles, small, small particles, stones, tiny groupings, tone, unknown minerals or drugs
made of metal	buckle, build, building, gray, hard, metal, metal tools, metallic, metallic tools, metals, shiny, silver, tools, use with hands
made of paper	books, card, classroom, collections, flatness, found in office, groups, has text, note-taking, office, paper, paper (colorful), papers, printed on, reading materials, rectangles, school, square, squares, stacks, striped, work
made of wood	brown, made of wood, natural, natural resources, orange, wood, wood-colored, yellow
marine	aquamarine life, aquatic activities, cruise, fish, in water, marine, nautical, ocean, outdoor, outside water, paradise, sea, ships, vacation, water

Table 4: Descriptions of abbreviation (A)

Abbreviation	Full descriptions
mineral/gem	beauty, clear minerals, crystal, earth-derived, gems, ice, in an artistic way, inspecting, intricate, jewelry, jewels, metallic, natural, natural minerals, prized, pure, rare, reflective, round, roundish, sharp, shiny, shinyness, sterile, translucent, valuable
musical instrument	control noise, hearing, instruments, listen, listening, loud, make noise, music, music instruments, musical, musical instruments, recreational instruments
object (bath)	bathroom, cleaning, essential everyday, gray, home (inside home), household items, hygiene, self-care, soap, toiletries, water, white
object (children)	baby, baby toys, child, child-like, children, dolls, toys, young, youth
object (house)	bland to colorful, chairs, cloth, common in household, everyday household, flat, furniture, house, house essentials, house surfaces, household commonness, household furniture, in house, living furniture, main component of room
object (medical)	health, health-concerning, hospital, hygiene, injury, medical, medical instruments, medical supplies, medicine, sick, to good health, unhealthy, water, wellness
object (small)	?, amish, appealing, candles, circular, color? Unsure, colorful, covers, cylinders, cylindrical, flat fat cylinders, hands-on, jewelry, lids, saving, shape, similar-shaped, things you grab, twine, unknown, yellow
outdoors	backyard, blue, brown colored, columns, common in outdoor, dirt, garden, landscape, man-made, monuments in nature, natural, nature, not colorful, outdoor objects, outdoorsy, park, pavement, pristine, public, quiet, rocks, rough, rows, scenery, separaters, stand on their own, statues, stone, tools, wood, woodsy, yard
permeable	can pass through, dot patterned, grates, holed, knit, little holes, mesh, metal, net, nets, octagonal, pattern, patterned, patterns, patterns (holes), repeated patterns, repeating, repeating patterns, repetitive, shiny, silver, small, strainer
plant	green, green leaves, green plants, green plants and herbs, greenery, greenness, greens, grow, natural, nature, plant-like, plants, things that grow from earth, vegetable, vegetables
round	artistic, ball, balls, circular, circular and colorful, circus, contrasting circles, fruits, kid, pictures, round, rund, shape, spherical
sport (ball)	athletic, ball toys, balls) to less active, competition, recreation, round, sport, sports, sports (active, sporty
string-like	amount of rigid ends, confetti, different shapes, elongated, hay, high-density, knots, lines, lines jutting out, long, long things mashed up, look like sticks, mesh, netting, patterns, prickly, protruding, repeating in an ordered way, rope, ropes, skinny, spiky, stacks, strings, stringy, symmetrical, tangled, twirled around
structure	amusement, antenna, big, caged, common to humans, complete, disordered, electrical, elongated, enclosures, found outside, grand, high, high in air, industrial, ladder, large, multiple cylinders, multiple similar things, narrow, outdoor, part of circle, shapes, stacks, structural, tall, things that go up, things that hang, trapped, wires
textured	appealing, carpets, flat coverings, fractals, lay flat, mesh, pattern, patterned, patterns, pieces, rectangles, repeating shapes, repetitive, rugs, sheets, small repeating patterns, squares, textured
tool	elongated, hand tools, household, jagged, long, pointy tools, pole, saws, scrape, sharp, sharp tools, small instruments, straight, tools, use with hands, utility, wedges
valuable/old	English royalty, antiquity, bottles, bronze, fine things, gaudiness, gold, high-class, high-quality, history, important, jewelry, jewels, monarchy, old, ornaments, precious metals, pristine, royal, royalty, shiny, silver, trophy, valueable
vehicle	can be moved, car, cars, complex vehicles, construction, efficiency of transportation, fast, ground motorized vehicles, machinery, mobility, move, speed of movement, transportation, transportation vehicles, travel with, truck, vehicles, wheeled vehicles, wheels
weapon	black, danger, dangerous, equipment, masculine, military, negatively-associated, ornaments, risky, self-defense, somber, violence, violent, war, war-like, weaponry, weapons
wearable	accessories, blue, can wear or carry or put on, clothes, clothing, cotton clothing, covering body, shirt, things to wear, things you wear, touch body, touch person, utilities, warm clothes, wearable
wheeled	able to hold, bicycle, caged, can sit on, can stand or drive, carrier, chair, destinations, holds objects, light movement, mobility, motion, move someone, playful, round, thing, things with wheels, trapped, wagon, wheel, wheeled-structures, wheels

Table 5: Descriptions of abbreviation (B)

Components of SPoSE dimension approximation

The following consecutive Table 6–9 show the component information of each SPoSE dimension. The percentage is calculated based on the portion $w_i \cdot \|O_{:,i}\|$, where w_i is the regression coefficient corresponding to our i-th embedding dimension. The right column shows the dimension-wise affordances.

SPoSE Labels	Components	Affordances (Top Ten Ranked Verbs)
animal	Dim 1 (14.16%):	spot, observe, photograph, find, sight, watch, note, kill, chase, hunt
	Dim 3 (13.73%):	hunt, kill, stalk, cull, chase, sight, exterminate, butcher, shoot, spot
	Dim 19 (7.93%):	pasture, slaughter, milk, herd, baa, bleat, fatten, graze, butcher, immolate
	Dim 23 (7.43%):	coach, lead, captain, rejoin, join, beat, fight, kill, face, battle
	Dim 33 (6.15%):	call, introduce, purchase, own, buy, choose, find, sell, bring, order
	Dim 46 (5.90%):	declaw, worm, leash, euthanize, cage, crate, emaciate, groom, mummify, frolic
food	Dim 16 (24.62%):	eat, munch, serve, bake, scoff, consume, cook, devour, prepare, taste
	Dim 33 (11.85%):	call, introduce, purchase, own, buy, choose, find, sell, bring, order
	Dim 2 (10.45%):	boil, blanch, cook, steam, drain, fry, sow, add, overcook, serve
	Dim 31 (8.52%):	cultivate, grow, plant, prune, propagate, gather, graft, dry, pick, pluck
	Dim 51 (7.36%):	sprinkle, mix, add, scatter, stir, toast, chop, sift, combine, taste
	Dim 64 (6.88%):	slice, peel, chop, grate, dice, quarter, mash, grow, cultivate, saute
	Dim 58 (6.65%):	spoon, thicken, simmer, stir, season, pour, spice, drizzle, reheat, taste
	Dim 6 (6.61%):	drink, sip, swig, sup, quaff, imbibe, guzzle, brew, swill, gulp
wearable	Dim 9 (54.91%):	wear, don, unbutton, button, match, knit, model, embroider, doff, rip
	Dim 25 (22.21%):	weave, drape, unfold, embroider, wave, fold, sew, wrap, hang, unfurl
	Dim 10 (12.97%):	unfasten, fasten, undo, tighten, buckle, unclip, unbuckle, loosen, loose, slacken
plant	Dim 31 (20.87%):	cultivate, grow, plant, prune, propagate, gather, graft, dry, pick, pluck
	Dim 2 (14.62%):	boil, blanch, cook, steam, drain, fry, sow, add, overcook, serve
	Dim 30 (12.20%):	grass, scarify, mow, smother, uproot, crop, water, plow, plough, mulch
	Dim 51 (10.02%):	sprinkle, mix, add, scatter, stir, toast, chop, sift, combine, taste
	Dim 64 (8.43%):	slice, peel, chop, grate, dice, quarter, mash, grow, cultivate, saute
	Dim 33 (5.32%):	call, introduce, purchase, own, buy, choose, find, sell, bring, order
vehicle	Dim 7 (67.90%):	park, drive, hire, crash, rent, commandeer, board, shunt, garage, equip
	Dim 59 (19.81%):	berth, capsize, drydock, moor, beach, charter, sail, board, crew, caulk
made of metal	Dim 66 (9.03%):	invent, manufacture, develop, design, introduce, employ, operate, buy, purchase, utilize
	Dim 5 (8.90%):	wield, sharpen, grab, hold, swing, carry, hand, grasp, thrust, clutch
	Dim 28 (8.78%):	unscrew, screw, slacken, weld, fit, disengage, lock, solder, tighten, undo
	Dim 27 (6.81%):	connect, disconnect, plug, wire, integrate, install, activate, incorporate, operate, mount
	Dim 24 (5.53%):	play, amplify, hear, learn, sound, perform, chime, study, tinkle, sing
	Dim 56 (5.33%):	mount, position, rotate, fit, incorporate, attach, move, lower, utilize, tilt
electronic device	Dim 27 (38.06%):	connect, disconnect, plug, wire, integrate, install, activate, incorporate, operate, mount
	Dim 66 (15.88%):	invent, manufacture, develop, design, introduce, employ, operate, buy, purchase, utilize
	Dim 33 (8.51%):	call, introduce, purchase, own, buy, choose, find, sell, bring, order
	Dim 24 (7.45%):	play, amplify, hear, learn, sound, perform, chime, study, tinkle, sing
	Dim 17 (5.44%):	print, photocopy, annotate, archive, scan, circulate, paste, forward, upload, mail
body/clothing	Dim 9 (24.62%):	wear, don, unbutton, button, match, knit, model, embroider, doff, rip
	Dim 4 (23.83%):	sprain, hyperextend, flex, bandage, fracture, injure, bruise, hurt, bend, rest
	Dim 48 (14.44%):	freckle, moisturize, spritz, wrinkle, redden, dehair, crinkle, rabbit, deflesh, exfoliate
	Dim 41 (9.36%):	cloister, unionize, pillory, ululate, intermarry, natter, famish, murmur, lynch, truncheon
	Dim 61 (8.51%):	ulcerate, distend, bare, zig, undulate, unhinge, decompress, sicken, pat, repopulate
musical instrument	Dim 24 (68.63%):	play, amplify, hear, learn, sound, perform, chime, study, tinkle, sing
	Dim 34 (8.44%):	honk, toot, hoot, beep, tootle, blare, whoop, sound, peep, journey
tool	Dim 5 (56.13%):	wield, sharpen, grab, hold, swing, carry, hand, grasp, thrust, clutch
	Dim 53 (9.57%):	moo, pomade, tweeze, cringe, gel, dehair, comb, shampoo, unfix, shingle
	Dim 69 (8.37%):	moisten, wet, dip, soak, rinse, dab, reuse, saturate, dampen, wipe
	Dim 66 (7.35%):	invent, manufacture, develop, design, introduce, employ, operate, buy, purchase, utilize
drink	Dim 6 (35.50%):	drink, sip, swig, sup, quaff, imbibe, guzzle, brew, swill, gulp
	Dim 13 (16.63%):	empty, fill, clean, clutter, line, tip, overturn, replenish, label, overload
	Dim 47 (14.17%):	formulate, squirt, smear, dilute, apply, evaporate, inject, dissolve, drip, dab
	Dim 58 (7.29%):	spoon, thicken, simmer, stir, season, pour, spice, drizzle, reheat, taste
	Dim 33 (6.57%):	call, introduce, purchase, own, buy, choose, find, sell, bring, order

Table 6: Components of SPoSE dimension approximation (A)

SPoSE Labels	Components	Affordances (Top Ten Ranked Verbs)
flammable	Dim 38 (56.31%):	light, extinguish, rekindle, kindle, ignite, douse, fuel, flicker, dim, flare
	Dim 54 (13.52%):	congeal, splutter, spew, solidify, gush, crystallize, extrude, ooze, emplace, belch
	Dim 21 (13.38%):	fire, lob, detonate, explode, hurl, discharge, throw, shoot, deflect, arm
can tie	Dim 10 (40.80%):	unfasten, fasten, undo, tighten, buckle, unclip, unbuckle, loosen, loose, slacken
	Dim 22 (31.45%):	coil, soldier, sever, thread, loop, splice, braid, wind, string, solder
bug	Dim 50 (26.87%):	horsewhip, slink, smirk, war, babysit, wriggle, battle, raft, chuckle, burrow
	Dim 1 (21.61%):	spot, observe, photograph, find, sight, watch, note, kill, chase, hunt
	Dim 35 (16.40%):	eradicate, deter, combat, exterminate, kill, prevent, eliminate, fight, discourage, swat
	Dim 68 (15.52%):	dicker, swat, zing, snipe, scrabble, fish, bridle, radio, blitz, coincide
granulated	Dim 65 (12.86%):	scuffle, tramp, shovel, whoosh, heap, accumulate, compact, sift, plough, shift
	Dim 57 (12.73%):	quarry, indurate, mine, vitrify, vein, leach, petrify, deposit, extract, mantle
	Dim 51 (10.34%):	sprinkle, mix, add, scatter, stir, toast, chop, sift, combine, taste
	Dim 37 (6.57%):	ossify, deflesh, calcify, fossilize, masticate, strengthen, jumble, bleach, fracture, gnash
	Dim 11 (5.69%):	steal, recover, retrieve, pawn, hide, smuggle, discover, snatch, possess, find
	Dim 49 (5.49%):	meow, fissure, furrow, burble, decoct, bawl, exfoliate, macerate, gnaw, speckle
	Dim 54 (5.40%):	congeal, splutter, spew, solidify, gush, crystallize, extrude, ooze, emplace, belch
weapon	Dim 21 (36.91%):	fire, lob, detonate, explode, hurl, discharge, throw, shoot, deflect, arm
	Dim 5 (23.89%):	wield, sharpen, grab, hold, swing, carry, hand, grasp, thrust, clutch
	Dim 10 (8.57%):	unfasten, fasten, undo, tighten, buckle, unclip, unbuckle, loosen, loose, slacken
	Dim 37 (7.13%):	ossify, deflesh, calcify, fossilize, masticate, strengthen, jumble, bleach, fracture, gnash
valuable/old	Dim 15 (24.90%):	carve, inscribe, desecrate, adorn, sculpt, flank, surmount, decorate, garland, erect
	Dim 11 (20.82%):	steal, recover, retrieve, pawn, hide, smuggle, discover, snatch, possess, find
	Dim 24 (18.59%):	play, amplify, hear, learn, sound, perform, chime, study, tinkle, sing
	Dim 9 (7.43%):	wear, don, unbutton, button, match, knit, model, embroider, doff, rip
	Dim 25 (6.28%):	weave, drape, unfold, embroider, wave, fold, sew, wrap, hang, unfurl
	Dim 38 (5.13%):	light, extinguish, rekindle, kindle, ignite, douse, fuel, flicker, dim, flare
object (house)	Dim 13 (18.33%):	empty, fill, clean, clutter, line, tip, overturn, replenish, label, overload
	Dim 25 (12.61%):	weave, drape, unfold, embroider, wave, fold, sew, wrap, hang, unfurl
	Dim 17 (7.13%):	print, photocopy, annotate, archive, scan, circulate, paste, forward, upload, mail
	Dim 12 (7.10%):	construct, erect, install, build, dismantle, situate, demolish, house, rebuild, repair
	Dim 63 (6.78%):	burl, lignify, distemper, stucco, paper, mottle, clam, plaster, crayon, whitewash
	Dim 27 (6.56%):	connect, disconnect, plug, wire, integrate, install, activate, incorporate, operate, mount
	Dim 67 (5.54%):	reorganize, federate, quake, nationalize, feud, staff, merge, privatize, amalgamate, inundate
	Dim 14 (5.45%):	yowl, toddle, mosey, totter, wheeze, sprint, troop, sledge, saunter, scamper
	Dim 28 (5.39%):	unscrew, screw, slacken, weld, fit, disengage, lock, solder, tighten, undo
aeronautic	Dim 1 (19.40%):	spot, observe, photograph, find, sight, watch, note, kill, chase, hunt
	Dim 59 (10.00%):	berth, capsize, drydock, moor, beach, charter, sail, board, crew, caulk
	Dim 53 (9.96%):	moo, pomade, tweeze, cringe, gel, dehair, comb, shampoo, unfix, shingle
	Dim 26 (5.89%):	lour, taxi, roil, scud, dissipate, eddy, ionize, tinge, decentralize, crimson
	Dim 42 (5.85%):	westernize, conflict, depose, neighbor, oust, encumber, assassinate, oppose, beard, wed
	Dim 15 (5.66%):	carve, inscribe, desecrate, adorn, sculpt, flank, surmount, decorate, garland, erect
	Dim 12 (5.42%):	construct, erect, install, build, dismantle, situate, demolish, house, rebuild, repair
textured	Dim 25 (24.35%):	weave, drape, unfold, embroider, wave, fold, sew, wrap, hang, unfurl
	Dim 65 (17.53%):	scuffle, tramp, shovel, whoosh, heap, accumulate, compact, sift, plough, shift
	Dim 57 (9.76%):	quarry, indurate, mine, vitrify, vein, leach, petrify, deposit, extract, mantle
	Dim 17 (8.36%):	print, photocopy, annotate, archive, scan, circulate, paste, forward, upload, mail
	Dim 63 (8.04%):	burl, lignify, distemper, stucco, paper, mottle, clam, plaster, crayon, whitewash
	Dim 70 (6.56%):	honeycomb, etch, handpaint, fabricate, emboss, weld, bond, distort, overlap, glue
	Dim 39 (6.16%):	block, obstruct, clog, flood, seal, dam, choke, swoosh, repair, widen
outdoors	Dim 36 (10.26%):	clamber, climb, descend, leap, recoil, channelize, clomp, ascend, jump, step
	Dim 12 (9.34%):	construct, erect, install, build, dismantle, situate, demolish, house, rebuild, repair
	Dim 8 (8.74%):	unload, lug, load, transport, wheel, stow, haul, offload, carry, dump
	Dim 30 (8.12%):	grass, scarify, mow, smother, uproot, crop, water, plow, plough, mulch
	Dim 5 (7.21%):	wield, sharpen, grab, hold, swing, carry, hand, grasp, thrust, clutch
	Dim 15 (6.01%):	carve, inscribe, desecrate, adorn, sculpt, flank, surmount, decorate, garland, erect
	Dim 39 (5.83%):	block, obstruct, clog, flood, seal, dam, choke, swoosh, repair, widen
	Dim 65 (5.19%):	scuffle, tramp, shovel, whoosh, heap, accumulate, compact, sift, plough, shift
body part	Dim 4 (36.06%):	sprain, hyperextend, flex, bandage, fracture, injure, bruise, hurt, bend, rest
	Dim 36 (8.40%):	clamber, climb, descend, leap, recoil, channelize, clomp, ascend, jump, step
	Dim 20 (7.92%):	ingrain, strangulate, punt, glove, fumble, vulcanize, bunt, slog, bat, loft
	Dim 37 (7.44%):	ossify, deflesh, calcify, fossilize, masticate, strengthen, jumble, bleach, fracture, gnash
	Dim 29 (7.11%):	gabble, loll, trill, bridle, lisp, bifurcate, manacle, slobber, ulcerate, cluck
	Dim 10 (6.10%):	unfasten, fasten, undo, tighten, buckle, unclip, unbuckle, loosen, loose, slacken
	Dim 69 (5.79%):	moisten, wet, dip, soak, rinse, dab, reuse, saturate, dampen, wipe
	Dim 9 (5.43%):	wear, don, unbutton, button, match, knit, model, embroider, doff, rip
made of paper	Dim 17 (33.19%):	print, photocopy, annotate, archive, scan, circulate, paste, forward, upload, mail
	Dim 8 (8.07%):	unload, lug, load, transport, wheel, stow, haul, offload, carry, dump
	Dim 66 (7.31%):	invent, manufacture, develop, design, introduce, employ, operate, buy, purchase, utilize
	Dim 11 (5.86%):	steal, recover, retrieve, pawn, hide, smuggle, discover, snatch, possess, find
	Dim 33 (5.40%):	call, introduce, purchase, own, buy, choose, find, sell, bring, order
	Dim 24 (5.36%):	play, amplify, hear, learn, sound, perform, chime, study, tinkle, sing

Table 7: Components of SPoSE dimension approximation (B)

SPoSE Labels	Components	Affordances (Top Ten Ranked Verbs)
round	Dim 31 (22.01%):	cultivate, grow, plant, prune, propagate, gather, graft, dry, pick, pluck
	Dim 64 (21.29%):	slice, peel, chop, grate, dice, quarter, mash, grow, cultivate, saute
	Dim 20 (17.28%):	ingrain, strangulate, punt, glove, fumble, vulcanize, bunt, slog, bat, loft
	Dim 62 (13.90%):	throb, fluoresce, zest, plash, glow, pulsate, pale, ream, redden, sequester
	Dim 57 (11.01%):	quarry, indurate, mine, vitrify, vein, leach, petrify, deposit, extract, mantle
made of wood	Dim 15 (8.76%):	carve, inscribe, desecrate, adorn, sculpt, flank, surmount, decorate, garland, erect
	Dim 40 (8.45%):	deforest, silicify, glimmer, degenerate, carpet, sandpaper, rut, plank, gasify, inlay
	Dim 12 (6.67%):	construct, erect, install, build, dismantle, situate, demolish, house, rebuild, repair
	Dim 8 (6.54%):	unload, lug, load, transport, wheel, stow, haul, offload, carry, dump
	Dim 49 (5.69%):	meow, fissure, furrow, burble, decoct, bawl, exfoliate, macerate, gnaw, speckle
	Dim 24 (5.67%):	play, amplify, hear, learn, sound, perform, chime, study, tinkle, sing
	Dim 5 (5.44%):	wield, sharpen, grab, hold, swing, carry, hand, grasp, thrust, clutch
	Dim 14 (5.03%):	yowl, toddle, mosey, totter, wheeze, sprint, troop, sledge, saunter, scamper
marine	Dim 59 (32.26%):	berth, capsize, drydock, moor, beach, charter, sail, board, crew, caulk
	Dim 1 (16.76%):	spot, observe, photograph, find, sight, watch, note, kill, chase, hunt
	Dim 50 (12.20%):	horsewhip, slink, smirk, war, babysit, wriggle, battle, raft, chuckle, burrow
	Dim 65 (6.03%):	scuffle, tramp, shovel, whoosh, heap, accumulate, compact, sift, plough, shift
structure	Dim 12 (38.26%):	construct, erect, install, build, dismantle, situate, demolish, house, rebuild, repair
	Dim 24 (14.95%):	play, amplify, hear, learn, sound, perform, chime, study, tinkle, sing
	Dim 56 (9.93%):	mount, position, rotate, fit, incorporate, attach, move, lower, utilize, tilt
	Dim 36 (8.30%):	clamber, climb, descend, leap, recoil, channelize, clomp, ascend, jump, step
	Dim 43 (7.04%):	determine, ascertain, desalt, increase, broaden, even, decrease, underline, plot, express
	Dim 10 (5.34%):	unfasten, fasten, undo, tighten, buckle, unclip, unbuckle, loosen, loose, slacken
mineral/gem	Dim 11 (33.55%):	steal, recover, retrieve, pawn, hide, smuggle, discover, snatch, possess, find
	Dim 57 (16.42%):	quarry, indurate, mine, vitrify, vein, leach, petrify, deposit, extract, mantle
	Dim 62 (12.45%):	throb, fluoresce, zest, plash, glow, pulsate, pale, ream, redden, sequester
	Dim 70 (10.53%):	honeycomb, etch, handpaint, fabricate, emboss, weld, bond, distort, overlap, glue
	Dim 26 (8.77%):	lour, taxi, roil, scud, dissipate, eddy, ionize, tinge, decentralize, crimson
object (bath)	Dim 47 (20.52%):	formulate, squirt, smear, dilute, apply, evaporate, inject, dissolve, drip, dab
	Dim 69 (20.33%):	moisten, wet, dip, soak, rinse, dab, reuse, saturate, dampen, wipe
	Dim 13 (15.94%):	empty, fill, clean, clutter, line, tip, overturn, replenish, label, overload
	Dim 39 (8.70%):	block, obstruct, clog, flood, seal, dam, choke, swoosh, repair, widen
	Dim 66 (7.34%):	invent, manufacture, develop, design, introduce, employ, operate, buy, purchase, utilize
	Dim 41 (6.45%):	cloister, unionize, pillory, ululate, intermarry, natter, famish, murmur, lynch, truncheon
	Dim 33 (5.60%):	call, introduce, purchase, own, buy, choose, find, sell, bring, order
accessory (face)	Dim 48 (23.10%):	freckle, moisturize, spritz, wrinkle, redden, dehair, crinkle, rabbit, deflesh, exfoliate
	Dim 9 (20.15%):	wear, don, unbutton, button, match, knit, model, embroider, doff, rip
	Dim 53 (17.05%):	moo, pomade, tweeze, cringe, gel, dehair, comb, shampoo, unfix, shingle
	Dim 44 (12.44%):	remonstrate, conk, room, prattle, billet, feminize, swaddle, waken, stutter, shush
	Dim 41 (6.75%):	cloister, unionize, pillory, ululate, intermarry, natter, famish, murmur, lynch, truncheon
	Dim 4 (6.21%):	sprain, hyperextend, flex, bandage, fracture, injure, bruise, hurt, bend, rest
elongated	Dim 5 (27.30%):	wield, sharpen, grab, hold, swing, carry, hand, grasp, thrust, clutch
	Dim 22 (13.68%):	coil, soldier, sever, thread, loop, splice, braid, wind, string, solder
	Dim 24 (10.69%):	play, amplify, hear, learn, sound, perform, chime, study, tinkle, sing
	Dim 56 (10.03%):	mount, position, rotate, fit, incorporate, attach, move, lower, utilize, tilt
	Dim 55 (8.01%):	twine, spiral, interlace, depressurize, creep, tussle, clothe, festoon, burr, misdirect
	Dim 38 (6.39%):	light, extinguish, rekindle, kindle, ignite, douse, fuel, flicker, dim, flare
	Dim 21 (5.48%):	fire, lob, detonate, explode, hurl, discharge, throw, shoot, deflect, arm
container	Dim 13 (40.45%):	empty, fill, clean, clutter, line, tip, overturn, replenish, label, overload
	Dim 8 (28.09%):	unload, lug, load, transport, wheel, stow, haul, offload, carry, dump
	Dim 67 (17.82%):	reorganize, federate, quake, nationalize, feud, staff, merge, privatize, amalgamate, inundate
accessory (beauty)	Dim 11 (46.04%):	steal, recover, retrieve, pawn, hide, smuggle, discover, snatch, possess, find
	Dim 9 (26.14%):	wear, don, unbutton, button, match, knit, model, embroider, doff, rip
	Dim 31 (8.79%):	cultivate, grow, plant, prune, propagate, gather, graft, dry, pick, pluck
wheeled	Dim 7 (39.07%):	park, drive, hire, crash, rent, commandeer, board, shunt, garage, equip
	Dim 8 (22.11%):	unload, lug, load, transport, wheel, stow, haul, offload, carry, dump
	Dim 58 (7.59%):	spoon, thicken, simmer, stir, season, pour, spice, drizzle, reheat, taste
	Dim 56 (7.53%):	mount, position, rotate, fit, incorporate, attach, move, lower, utilize, tilt
	Dim 14 (7.43%):	yowl, toddle, mosey, totter, wheeze, sprint, troop, sledge, saunter, scamper
string-like	Dim 22 (26.52%):	coil, soldier, sever, thread, loop, splice, braid, wind, string, solder
	Dim 30 (20.06%):	grass, scarify, mow, smother, uproot, crop, water, plow, plough, mulch
	Dim 53 (17.38%):	moo, pomade, tweeze, cringe, gel, dehair, comb, shampoo, unfix, shingle
	Dim 25 (14.20%):	weave, drape, unfold, embroider, wave, fold, sew, wrap, hang, unfurl
	Dim 57 (11.80%):	quarry, indurate, mine, vitrify, vein, leach, petrify, deposit, extract, mantle
agglomerate	Dim 70 (13.02%):	honeycomb, etch, handpaint, fabricate, emboss, weld, bond, distort, overlap, glue
	Dim 69 (11.32%):	moisten, wet, dip, soak, rinse, dab, reuse, saturate, dampen, wipe
	Dim 11 (10.16%):	steal, recover, retrieve, pawn, hide, smuggle, discover, snatch, possess, find
	Dim 18 (9.50%):	winnow, mill, parboil, reap, grind, dehusk, export, contaminate, stockpile, sift
	Dim 40 (7.00%):	deforest, silicify, glimmer, degenerate, carpet, sandpaper, rut, plank, gasify, inlay
	Dim 8 (5.96%):	unload, lug, load, transport, wheel, stow, haul, offload, carry, dump
	Dim 22 (5.88%):	coil, soldier, sever, thread, loop, splice, braid, wind, string, solder
	Dim 31 (5.01%):	cultivate, grow, plant, prune, propagate, gather, graft, dry, pick, pluck

Table 8: Components of SPoSE dimension approximation (C)

SPoSE Labels	Components	Affordances (Top Ten Ranked Verbs)
object (children)	Dim 44 (32.67%):	remonstrate, conk, room, prattle, billet, feminize, swaddle, waken, stutter, shush
	Dim 17 (16.30%):	print, photocopy, annotate, archive, scan, circulate, paste, forward, upload, mail
	Dim 33 (9.05%):	call, introduce, purchase, own, buy, choose, find, sell, bring, order
	Dim 24 (8.03%):	play, amplify, hear, learn, sound, perform, chime, study, tinkle, sing
	Dim 15 (7.61%):	carve, inscribe, desecrate, adorn, sculpt, flank, surmount, decorate, garland, erect
	Dim 11 (7.33%):	steal, recover, retrieve, pawn, hide, smuggle, discover, snatch, possess, find
sport (ball)	Dim 20 (17.80%):	ingrain, strangulate, punt, glove, fumble, vulcanize, bunt, slog, bat, loft
	Dim 24 (11.97%):	play, amplify, hear, learn, sound, perform, chime, study, tinkle, sing
	Dim 56 (10.28%):	mount, position, rotate, fit, incorporate, attach, move, lower, utilize, tilt
	Dim 9 (8.30%):	wear, don, unbutton, button, match, knit, model, embroider, doff, rip
	Dim 7 (7.77%):	park, drive, hire, crash, rent, commandeer, board, shunt, garage, equip
	Dim 10 (6.64%):	unfasten, fasten, undo, tighten, buckle, unclip, unbuckle, loosen, loose, slacken
	Dim 44 (5.55%):	remonstrate, conk, room, prattle, billet, feminize, swaddle, waken, stutter, shush
	Dim 59 (5.43%):	berth, capsize, drydock, moor, beach, charter, sail, board, crew, caulk
appliance/machine	Dim 8 (17.50%):	unload, lug, load, transport, wheel, stow, haul, offload, carry, dump
	Dim 27 (16.67%):	connect, disconnect, plug, wire, integrate, install, activate, incorporate, operate, mount
	Dim 39 (10.77%):	block, obstruct, clog, flood, seal, dam, choke, swoosh, repair, widen
	Dim 70 (8.13%):	honeycomb, etch, handpaint, fabricate, emboss, weld, bond, distort, overlap, glue
	Dim 7 (6.92%):	park, drive, hire, crash, rent, commandeer, board, shunt, garage, equip
	Dim 64 (5.19%):	slice, peel, chop, grate, dice, quarter, mash, grow, cultivate, saute
	Dim 13 (5.03%):	empty, fill, clean, clutter, line, tip, overturn, replenish, label, overload
flat surface	Dim 25 (36.96%):	weave, drape, unfold, embroider, wave, fold, sew, wrap, hang, unfurl
	Dim 69 (21.94%):	moisten, wet, dip, soak, rinse, dab, reuse, saturate, dampen, wipe
	Dim 53 (14.34%):	moo, pomade, tweeze, cringe, gel, dehair, comb, shampoo, unfix, shingle
	Dim 17 (13.88%):	print, photocopy, annotate, archive, scan, circulate, paste, forward, upload, mail
object (medical)	Dim 37 (17.99%):	ossify, deflesh, calcify, fossilize, masticate, strengthen, jumble, bleach, fracture, gnash
	Dim 11 (16.23%):	steal, recover, retrieve, pawn, hide, smuggle, discover, snatch, possess, find
	Dim 13 (15.14%):	empty, fill, clean, clutter, line, tip, overturn, replenish, label, overload
	Dim 66 (13.94%):	invent, manufacture, develop, design, introduce, employ, operate, buy, purchase, utilize
	Dim 29 (6.12%):	gabble, loll, trill, bridle, lisp, bifurcate, manacle, slobber, ulcerate, cluck
	Dim 69 (5.76%):	moisten, wet, dip, soak, rinse, dab, reuse, saturate, dampen, wipe
craft	Dim 25 (61.86%):	weave, drape, unfold, embroider, wave, fold, sew, wrap, hang, unfurl
	Dim 22 (19.73%):	coil, soldier, sever, thread, loop, splice, braid, wind, string, solder
circular	Dim 20 (25.55%):	ingrain, strangulate, punt, glove, fumble, vulcanize, bunt, slog, bat, loft
	Dim 58 (12.19%):	spoon, thicken, simmer, stir, season, pour, spice, drizzle, reheat, taste
	Dim 32 (9.88%):	motorcycle, voyage, zigzag, crisscross, roil, span, pulsate, repopulate, journey, bestride
	Dim 39 (9.76%):	block, obstruct, clog, flood, seal, dam, choke, swoosh, repair, widen
	Dim 56 (8.32%):	mount, position, rotate, fit, incorporate, attach, move, lower, utilize, tilt
	Dim 16 (8.26%):	eat, munch, serve, bake, scoff, consume, cook, devour, prepare, taste
	Dim 43 (7.65%):	determine, ascertain, desalt, increase, broaden, even, decrease, underline, plot, express
colorful pattern	Dim 25 (22.55%):	weave, drape, unfold, embroider, wave, fold, sew, wrap, hang, unfurl
	Dim 47 (17.84%):	formulate, squirt, smear, dilute, apply, evaporate, inject, dissolve, drip, dab
	Dim 43 (13.42%):	determine, ascertain, desalt, increase, broaden, even, decrease, underline, plot, express
	Dim 22 (13.26%):	coil, soldier, sever, thread, loop, splice, braid, wind, string, solder
	Dim 24 (10.89%):	play, amplify, hear, learn, sound, perform, chime, study, tinkle, sing
	Dim 17 (7.21%):	print, photocopy, annotate, archive, scan, circulate, paste, forward, upload, mail
permeable	Dim 56 (33.70%):	mount, position, rotate, fit, incorporate, attach, move, lower, utilize, tilt
	Dim 39 (29.94%):	block, obstruct, clog, flood, seal, dam, choke, swoosh, repair, widen
	Dim 43 (8.57%):	determine, ascertain, desalt, increase, broaden, even, decrease, underline, plot, express
	Dim 63 (7.81%):	burl, lignify, distemper, stucco, paper, mottle, clam, plaster, crayon, whitewash
accessory (black)	Dim 47 (24.09%):	formulate, squirt, smear, dilute, apply, evaporate, inject, dissolve, drip, dab
	Dim 9 (14.28%):	wear, don, unbutton, button, match, knit, model, embroider, doff, rip
	Dim 6 (13.96%):	drink, sip, swig, sup, quaff, imbibe, guzzle, brew, swill, gulp
	Dim 56 (13.17%):	mount, position, rotate, fit, incorporate, attach, move, lower, utilize, tilt
	Dim 11 (11.46%):	steal, recover, retrieve, pawn, hide, smuggle, discover, snatch, possess, find
	Dim 41 (8.99%):	cloister, unionize, pillory, ululate, intermarry, natter, famish, murmur, lynch, truncheon
object (small)	Dim 57 (21.67%):	quarry, indurate, mine, vitrify, vein, leach, petrify, deposit, extract, mantle
	Dim 38 (21.52%):	light, extinguish, rekindle, kindle, ignite, douse, fuel, flicker, dim, flare
	Dim 22 (19.43%):	coil, soldier, sever, thread, loop, splice, braid, wind, string, solder
	Dim 67 (12.81%):	reorganize, federate, quake, nationalize, feud, staff, merge, privatize, amalgamate, inundate
	Dim 15 (10.01%):	carve, inscribe, desecrate, adorn, sculpt, flank, surmount, decorate, garland, erect
degree of red	Dim 31 (31.07%):	cultivate, grow, plant, prune, propagate, gather, graft, dry, pick, pluck
	Dim 47 (13.79%):	formulate, squirt, smear, dilute, apply, evaporate, inject, dissolve, drip, dab
	Dim 54 (10.17%):	congeal, splutter, spew, solidify, gush, crystallize, extrude, ooze, emplace, belch
	Dim 24 (9.97%):	play, amplify, hear, learn, sound, perform, chime, study, tinkle, sing
	Dim 7 (7.33%):	park, drive, hire, crash, rent, commandeer, board, shunt, garage, equip
	Dim 10 (5.85%):	unfasten, fasten, undo, tighten, buckle, unclip, unbuckle, loosen, loose, slacken
furry	Dim 26 (24.94%):	lour, taxi, roil, scud, dissipate, eddy, ionize, tinge, decentralize, crimson
	Dim 37 (20.76%):	ossify, deflesh, calcify, fossilize, masticate, strengthen, jumble, bleach, fracture, gnash
	Dim 19 (19.32%):	pasture, slaughter, milk, herd, baa, bleat, fatten, graze, butcher, immolate
	Dim 53 (11.36%):	moo, pomade, tweeze, cringe, gel, dehair, comb, shampoo, unfix, shingle
	Dim 40 (8.65%):	deforest, silicify, glimmer, degenerate, carpet, sandpaper, rut, plank, gasify, inlay

Table 9: Components of SPoSE dimension approximation (D)